

MANSA AGGARWAL

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SUMMARY

Computer Science graduate focused on the intersection of AI/ML and embedded systems. Experience building real-time computer vision applications and WiFi-connected embedded hardware, with a track record of shipping complete, working projects end-to-end.

EDUCATION

B.Tech, Computer Science — M.E.R.I. College of Engineering & Technology (MDU)

PROJECTS

Gesture App | <https://github.com/mansatayal/gesture-app>

- Built a real-time hand gesture recognition app in Python using OpenCV, MediaPipe, and Pygame, supporting three interactive modes from a single webcam pipeline.
- Designed a gesture-controlled drawing canvas (Draw mode) with adjustable brush sizes, an eraser, and a timed color-selection system using a 7-swatch palette.
- Built a presentation controller (Count mode) that detects finger-count gestures across two hands simultaneously and maps them to slide navigation (next/previous/start/stop) via PyAutoGUI, with hold-based gesture confirmation and audio feedback.
- Optimized MediaPipe inference (model_complexity=0) for real-time performance and documented model limitations around hand orientation in the project README.

ESP32 Weather Dashboard | <https://github.com/mansatayal/esp32-weather-dashboard>

- Built a self-hosted IoT dashboard on the ESP32, reading live temperature and humidity from a DHT11 sensor and serving readings via an on-device WiFi web server.
- Handled end-to-end hardware bring-up: sensor wiring, firmware in Arduino C++, and driver/board configuration troubleshooting.

Traffic Light Controller with LED Matrix Countdown | <https://github.com/mansatayal/traffic-light-arduino>

- Built a traffic signal simulator in Arduino C++ with a synced 8x8 LED matrix countdown timer for each light state, including a custom digit bitmap font rendered via MAX7219 driver.

Machine Learning Coursework | <https://github.com/mansatayal/ml-coursework>

- Built and evaluated ML models across regression, classification, and deep learning tasks — including a CNN for image classification (~83% accuracy) and ANN-based multi-class classification — using Python, scikit-learn, and TensorFlow/Keras.

SKILLS

Languages: Python, C++

AI / ML: OpenCV, MediaPipe, TensorFlow, Keras, scikit-learn

Embedded / Hardware: Arduino, ESP32, OpenWrt (router firmware), sensor integration

Tools: Git, GitHub, VS Code, Pygame, PyAutoGUI

Coursework: Data Structures & Algorithms, Discrete Mathematics, Machine Learning